

SHARIF UNIVERSITY OF TECHNOLOGY

Department of Electrical Engineering

ELECTRONICS II

PROJECT 2

Design and Simulation of a Low-Distortion Power Amplifier

Folded-cascode input stage, active bias networks and complementary output drive for a 100 Ω load

Circuit Design and LTspice Simulation

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1. Design problem and requirements

The project concerns a low-frequency voltage/power amplifier able to source and sink substantial load current while remaining accurately linear. A small-signal differential input stage must create adequate open-loop gain; a complementary emitter-follower output stage supplies current to a 100 Ω load; and negative feedback sets the overall voltage gain. The initial topology is examined to identify lost voltage headroom and crossover/dead-zone effects. The redesign preserves the output principle while implementing a folded-cascode front end and transistor-based bias/reference circuits.

Table 1. Design targets and evaluation conditions.

Design constraint	Target
Supply voltage	VCC=+10 V, VEE=−10 V
Load resistance	100 Ω resistive
Closed-loop voltage gain	2 V/V
Symmetrical output swing	More than 16.6 V peak-to-peak
Total harmonic distortion	Less than 0.05% at 16.4 V peak-to-peak output
Quiescent power	Approximately 50 mW target
Number of transistors	Strictly less than 25
Devices	2N3904-type NPN, βF=200; 2N3906-type PNP, βF=150, as defined in LTspice
Additional test	Change both feedback resistors to 50 kΩ and examine waveform distortion

2. Reference circuit and its limitations

The reference architecture has a transistor-level differential input and voltage-gain path that drives a complementary output pair. The intermediate transistor Q6 serves as a buffer/driver, reducing the loading of the preceding gain node. A current sink I_B establishes the driver bias and helps sustain conduction in Q5. In the original reference topology, Q7 and Q8 deliver positive and negative load currents, while the feedback resistor network regulates the output voltage.

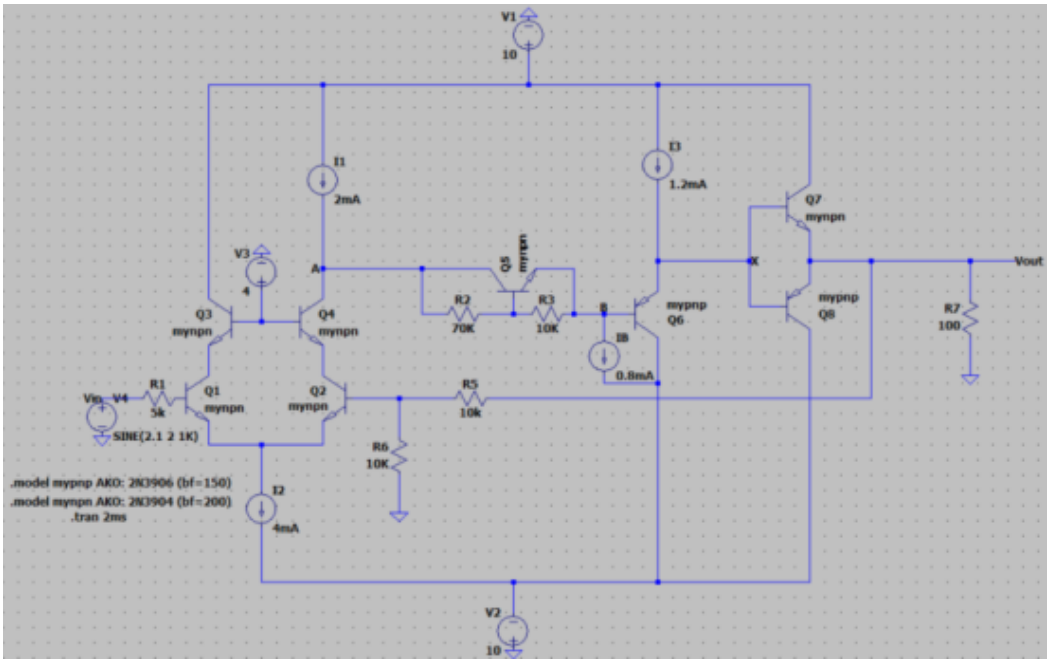


Figure 1. LTspice implementation of the reference topology used before the folded-cascode redesign.

2.1 Class-B dead zone and feedback

An unbiased complementary emitter-follower pair requires a finite base-emitter voltage before either device conducts substantially. Around zero-crossing, this leaves a transfer region with reduced output response. Negative feedback forces the preceding voltage-gain stage to develop additional drive and can largely suppress the output notch while the loop remains within its headroom and bandwidth. Feedback cannot recover swing lost when an internal driver saturates or when the supply-rail headroom is exhausted.

The source design records a bias-current adjustment around I_B=0.08 mA for the early topology. This is a design iteration, not a calibrated final-stage measurement. The initial simulation run with a nominal sinusoidal input produces V_{out} between approximately 0.081 and 7.884 V, indicating a strongly asymmetric response and limited headroom in that configuration.

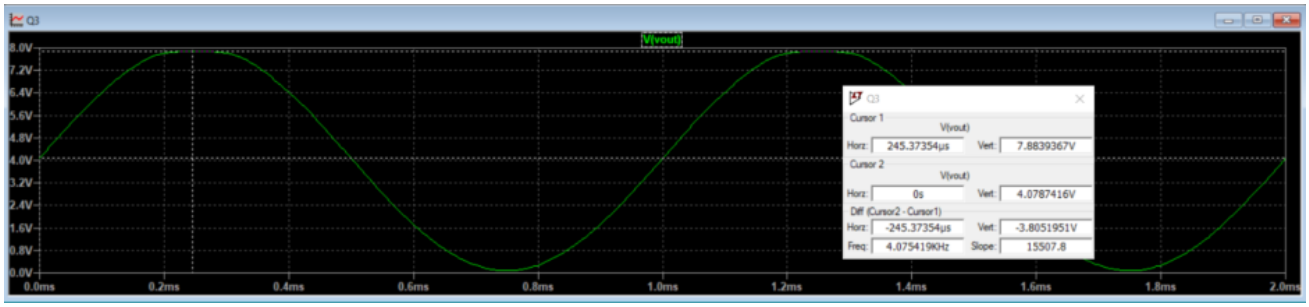


Figure 2. Reference-circuit output waveform showing limited, offset swing before redesign.

2.2 Why the first-stage voltage headroom matters

The output devices require adequate control voltage across their bases and emitters. If the upstream gain stage becomes limited by the collector-emitter saturation voltage, current-source compliance or a fixed bias drop, increasing the input merely shifts internal nodes rather than extending the load voltage. A folded-cascode front end separates the signal input common-mode requirements from the high-impedance gain node and makes it possible to choose favorable DC levels while maintaining gain.

2.3 Bias current, transistor operating points and input common-mode range

The initial differential input section splits a tail current between its two branches at a balanced zero-differential input; in first-order analysis, $I_{C1} \approx I_{C2} \approx I_{tail}/2$. The Q6 driver and its bias sink satisfy a node-current constraint of the form $I_B \approx I_{driver} + I_{bias\text{-}network}$, allowing adequate drive current while maintaining the preceding device in conduction. Exact current values depend on the BJT models and the chosen DC source settings and must be read from the matching operating point rather than an ideal β-only estimate.

The input common-mode range (ICMR) is determined at the upper end by the supply-side current-source and input-transistor compliance, and at the lower end by the tail-current-source compliance and available base-emitter drops. A useful headroom expression is $V_{CM,max} \approx VCC - V_{headroom,top} - V_{BE}$, with a corresponding $V_{CM,min} \approx VEE + V_{headroom,tail} + V_{BE}$. These are architecture-level estimates, not numerical limits obtained from an recorded common-mode sweep. The source package contains no stored full ICMR sweep for the selected initial configuration, so a numerical final ICMR is not asserted.

3. Folded-cascode redesign

The final design introduces a differential transconductance input followed by a folded current path and high-output-resistance active load. In a cascode device, the signal transistor sees a reduced voltage variation, which limits Miller feedback and can preserve voltage gain while improving high-frequency behavior. The exact bandwidth improvement cannot be quantified here without a matching frequency-domain comparison. The important function in this design is to shift the high-gain node to a bias range that better drives the complementary output stage.

Bias is established by current references and mirrored into the stages rather than by independent ideal current generators. The original redesign description identifies emitter-degenerated mirror branches, a β-helper arrangement for improved mirror accuracy, and self-biasing of the cascode transistors. An active load is also employed in place of a resistive collector load where higher incremental resistance is needed. These features reduce dependence on uncertain transistor current gain but do not eliminate model and temperature sensitivity.

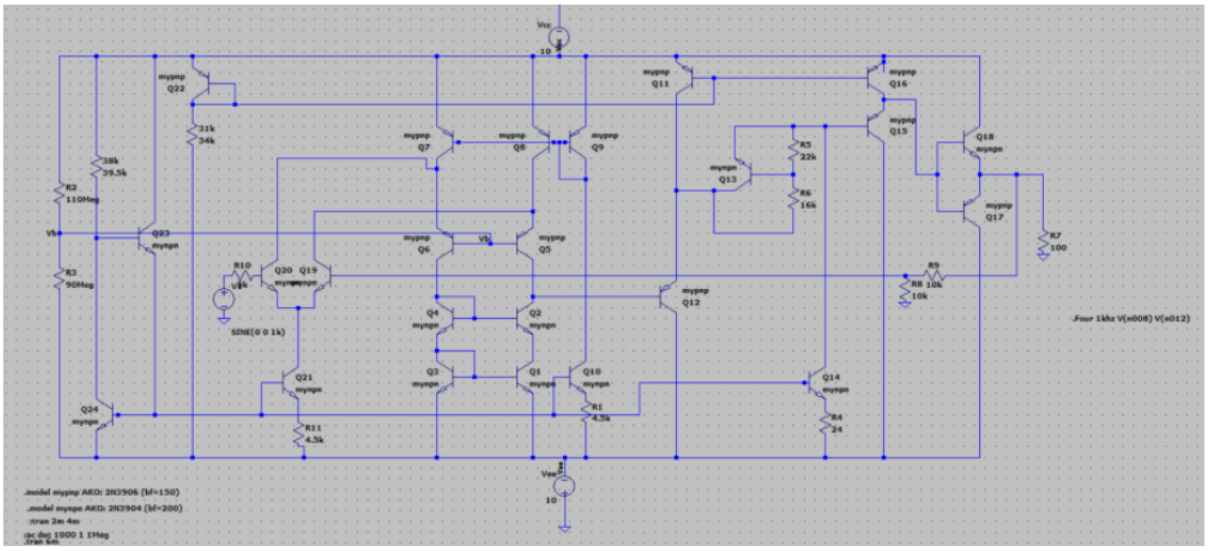


Figure 3. Final 24-transistor LTspice power amplifier (Q5.asc), combining bias mirrors, folded-cascode gain section, driver stage and complementary output pair.

3.1 Final circuit components and constraints

The final circuit schematic contains 24 BJT symbols, under the 25-transistor limit. It specifies 2N3904-derived NPN and 2N3906-derived PNP models with forward current gains of 200 and 150, respectively. The output load R7 is 100 Ω. Both feedback resistors R8 and R9 are 10 kΩ, consistent with a nominal non-inverting gain near 1+R9/R8=2 when the loop gain is large. The circuit also includes 3.5 kΩ, 16 kΩ and 22 kΩ reference/degeneration elements, selected as part of the recorded bias implementation.

The bias solution should be understood as a working simulation for the specified model and nominal temperature. Neither a corner sweep nor a temperature or component-tolerance analysis is recorded. Consequently, nominal gain, power and distortion must not be interpreted as process- or temperature-robust guarantees.

4. Simulation methodology

The principal final files is **Q5.asc**. It defines +10 V and −10 V sources and a 1 kHz input sine wave of **4.2 V peak amplitude**. The transient directive integrates the circuit for four milliseconds, with a requested output increment of two milliseconds; the .FOUR directive operates at 1 kHz on node V(n007), the loaded output. The associated .raw, .op.raw and .log files are available, enabling direct recovery of amplitude, zero-signal operating point and the harmonic table.

Table 2. Reproducible operating conditions from the final saved schematic.

Quantity	Simulation setup
Circuit file	Q5.asc
Input waveform	SINE(0 4.2 1k)
Supplies	±10 V
Load and nominal feedback	100 Ω; R8=R9=10 kΩ
Transient analysis	.tran 2m 4m
Fourier analysis	.four 1kHz V(n007)
Transistor models	mypnp AKO:2N3906 (βF=150); mynpn AKO:2N3904 (βF=200)
Conditions	Nominal LTspice model; no saved process or temperature sweep

5. Transient output swing and closed-loop gain

The output waveform is nearly sinusoidal for the 4.2 V-amplitude drive. On the last 1 ms interval of the stored transient simulation, its observed extrema are V_{max}=+8.4034 V and V_{min}=−8.3874 V. Therefore:

$$V_{out,pp} = V_{max} - V_{min} = 16.7908\ \text{V}\ .$$

The positive and negative peaks are close in magnitude. This run exceeds the minimum 16.6 Vpp symmetric-swing target. The Fourier fundamental amplitude is 8.395 V and the input fundamental is 4.2 V, so the 1 kHz fundamental gain is approximately 8.395 / 4.2 = **1.9988 V/V**, close to the requested gain of two.

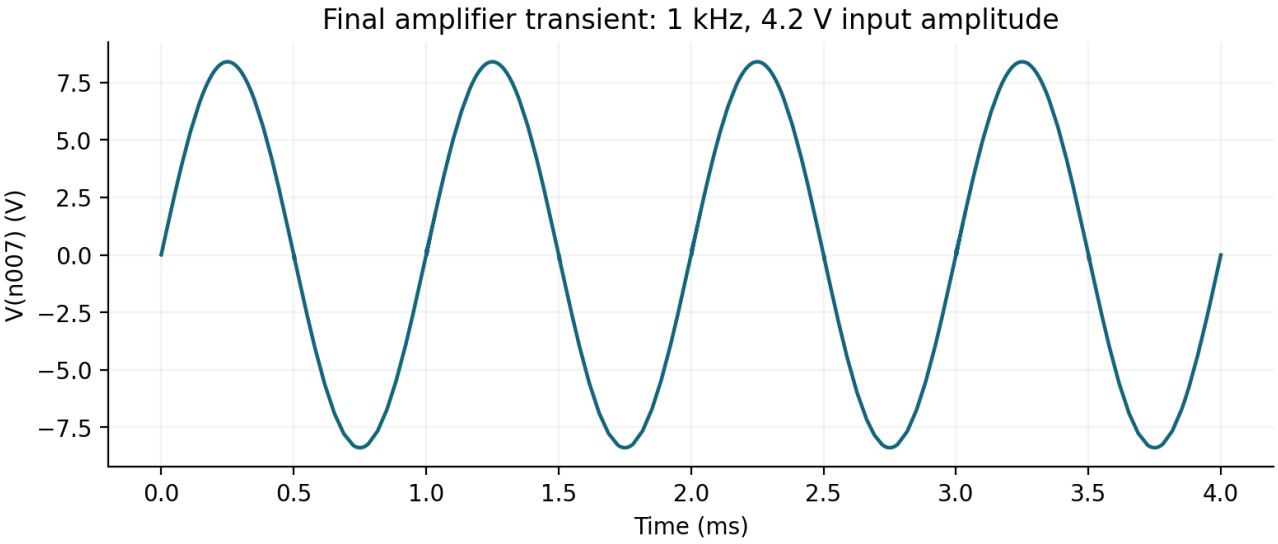


Figure 4. Loaded output waveform reconstructed from the saved Q5.raw transient data.

5.1 Load current and signal power

At a sinusoidal voltage peak near 8.395 V and a purely resistive 100 Ω load, the peak load current is approximately 84 mA. The corresponding sinusoidal average load power is $P_{load} \approx (8.395\ V)^2/(2\times 100\ \Omega) = \mathbf{0.352\ W}$. This is output power during the driven test and is not the quiescent dissipated power of the amplifier. A full-efficiency figure would require simultaneous average supply-power measurements during the same driven period.

6. Quiescent consumption

The final .op.raw file was saved with the input sinusoid at its zero-time value, giving the zero-signal quiescent operating point. The supply currents are I(VCC)=−2.35993 mA and I(VEE)=−2.28843 mA. Summing the power delivered by the ±10 V supplies gives

$$P_Q = 10|I(VCC)| + 10|I(VEE)| = 46.48\ mW\ .$$

This direct operating-point estimate is consistent with the approximately 45.8 mW level illustrated in the initial project report. The small difference is attributable to differing saved measurements or calculation conventions; the 46.48 mW result is the value recoverable from the saved numerical operating point.

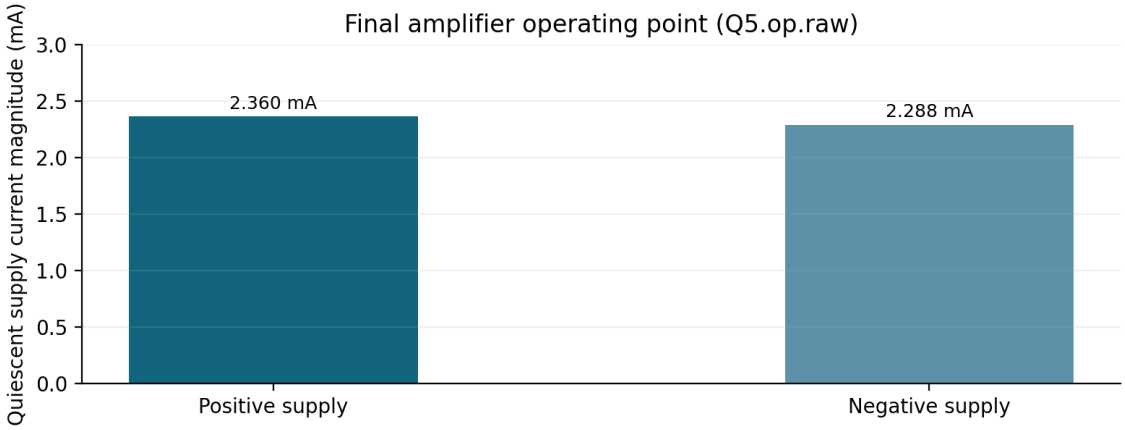


Figure 5. Magnitude of the two DC supply currents directly recovered from the final operating-point file (Q5.op.raw).

7. Harmonic distortion at high output level

Total harmonic distortion compares the root-sum-square of harmonics above the fundamental to the fundamental amplitude. In a voltage waveform with Fourier amplitudes V_n, the usual finite-harmonic definition is

$$\mathrm{THD} = 100 \times \sqrt{\left(\Sigma_{n\geq 2}\, V_n^{\,2}\right)} / V_1\ [\%]\ .$$

The LTspice .FOUR output for V(n007) reports a **1 kHz fundamental of 8.395 V** and **total harmonic distortion of 0.047880%** for the saved 4.2 V-amplitude input. The printed partial harmonic distortion for the nine listed components is 0.016008%; it is not interchangeable with LTspice's reported total THD, which includes additional harmonic content according to its computation settings.

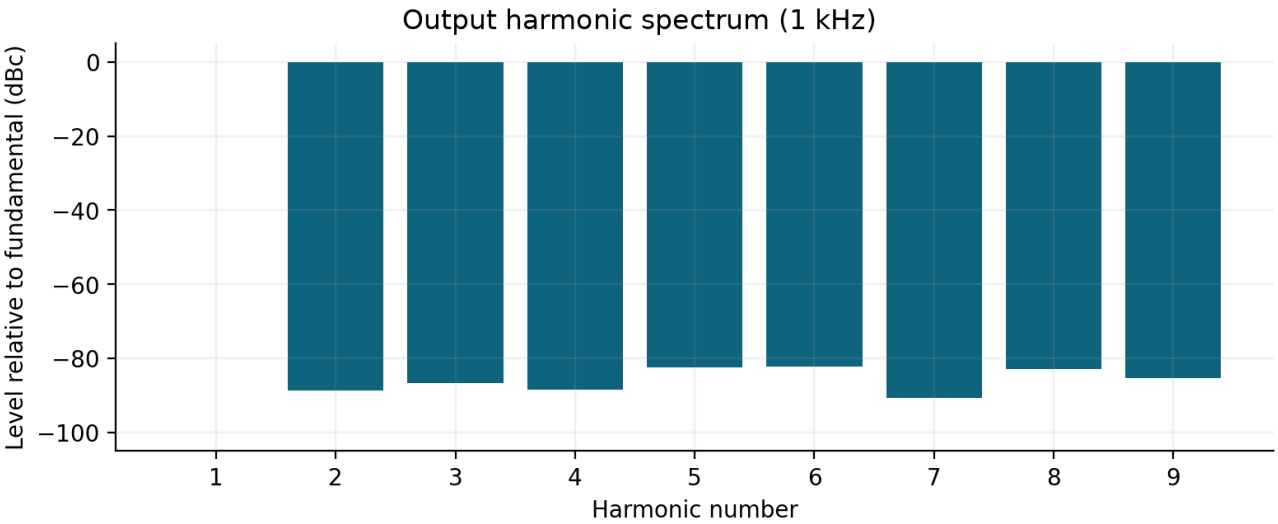


Figure 6. Harmonic amplitudes from the final Q5.log, normalized to the 1 kHz fundamental; levels are plotted in dBc.

Table 3. Selected FFT components parsed directly from Q5.log. Relative levels are calculated from the displayed Fourier amplitudes.

Harmonic	Frequency	Fourier amplitude (V)	Relative amplitude
1	1 kHz	8.395	0 dBc
2	2 kHz	3.080×10^{-4}	−88.7 dBc
3	3 kHz	3.859×10^{-4}	−86.8 dBc
5	5 kHz	6.359×10^{-4}	−82.4 dBc
6	6 kHz	6.472×10^{-4}	−82.3 dBc
8	8 kHz	6.078×10^{-4}	−82.8 dBc

A precision issue remains: the formal assignment requests THD at **exactly 16.4 Vpp**, whereas the reported 0.047880% THD was obtained with approximately 16.79 Vpp output. The stored log therefore establishes compliance at the measured 16.79 Vpp condition, not an independent numeric measurement at exactly 16.4 Vpp. Although the neighboring operating points may perform similarly, such an inference should not replace a dedicated 16.4 Vpp simulation.

8. Feedback-network sensitivity

The course experiment also requires setting both feedback resistors to 50 kΩ. Holding their ratio constant preserves the nominal ideal closed-loop gain, but it changes the absolute impedances seen by the input and feedback nodes. BJT input currents, bias mismatches and finite loop gain can then create DC shifts or increased distortion. Large feedback resistances can also change coupling to parasitic capacitances and frequency-domain stability.

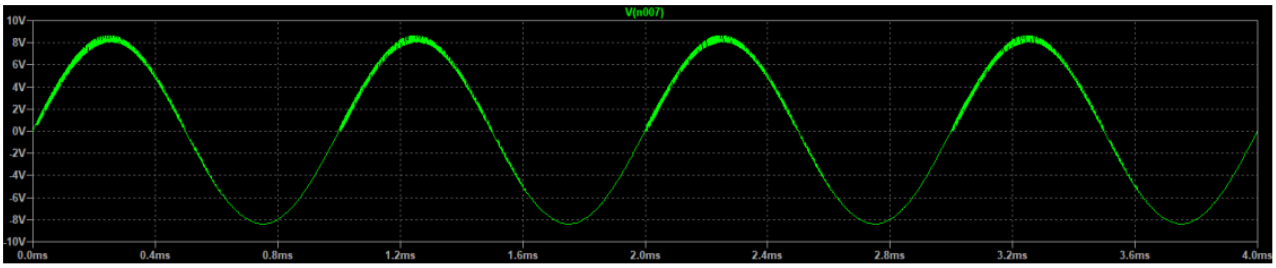


Figure 7. Transient output with the modified 50 kΩ feedback network. No corresponding RAW file is available for a numerical THD comparison.

The initial project report attributes the observed waveform deterioration to a loss of input-stage bias symmetry. That explanation is physically plausible, but the available simulation files do not include a controlled before/after operating-point and Fourier pair for the 50 kΩ configuration. The specific distortion mechanism cannot therefore be isolated quantitatively. The proposed diagnosis is testable by comparing differential-pair collector currents, internal saturation margins, output offset and THD while varying only the two feedback resistors.

9. Design verification

Table 4. Comparison with specified design targets.

Requirement	Simulation results	Status
Voltage gain ≈2	1 kHz Fourier fundamental 8.395 V with 4.2 V-amplitude input → 1.9988	Met in saved nominal run
Swing >16.6 Vpp	+8.4034 V / −8.3874 V → 16.7908 Vpp	Met in saved nominal run
Quiescent power ≈50 mW	46.48 mW from .op source currents	Consistent with target
THD <0.05% at 16.4 Vpp	0.047880% at approximately 16.79 Vpp	Met at measured amplitude; exact specified amplitude untested
Less than 25 BJTs	24 transistor symbols in final Q5.asc	Met
50 kΩ feedback test	Source-report image and qualitative explanation	Observed, not numerically reproduced
Folded-cascode redesign	Final .asc and design documentation	Implemented; no isolation of bandwidth benefit

10. Conclusion

At ±10 V and a 100 Ω load, the 24-transistor amplifier achieves a gain of 1.9988, 16.79 Vpp swing and 46.48 mW quiescent power. The stored THD is 0.047880% at 16.79 Vpp; compliance at exactly 16.4 Vpp has not been measured.